CS454 Midterm 2 Part A

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I. Questions

A. General Learning

1. In which of the following machine learning applications there is a supervisor and it tries to estimate a continuous entity from a set of possibilities?

a) Association
b) Classification
c) Regression
d) Unsupervised Learning
e) Reinforcement Learning

B. Classification

Given the following classification dataset $\mathcal{X}_1$. Answer question 2 accordingly.

$$\mathcal{X}_1 = \begin{bmatrix} & x_1 & x_2 & x_3 & C \\ \hline x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & 3 & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & green & no \\ x^{(5)} & 3 & 1 & blue & no \end{bmatrix}$$

2. How many features are discrete?

a) 1
b) 2
c) 3
d) 4
e) 5

C. Classification Metrics

Given the dataset in the questions above, the classification decisions for two classifiers are given below. Answer questions 3-4 accordingly.

$$\begin{bmatrix} Classifier_1 & Classifier_2 \\ \hline no & yes \\ no & no \\ yes & yes \\ yes & no \\ yes & yes \end{bmatrix}$$

3. What is the error rate of $Classifier_2$?

a) 0
b) 0.2
c) 0.8
d) 0.4
e) 1

4. If yes is the positive class, then what is the precision of $Classifier_1$?

a) 1 / 1
b) 1 / 2
c) 1 / 3
d) 3 / 4
e) 2 / 3

D. Regression Metrics

Given the dataset in the questions above, the regression decisions for two regressors are given below. Answer question 5 accordingly.

$$\begin{bmatrix} Regressor_1 & Regressor_2 \\ \hline 4.0 & 4.0 \\ 6.0 & 3.6 \\ 3.0 & 2.2 \\ 7.0 & 6.4 \end{bmatrix}$$

5. What is the mean square error of Regressor$_2$?

a) 0.01
b) 0.02
c) 0.03
d) 0.04
e) 0.05

E. K class vs. 2 class classification problems

6. If we have a $K = 4$ class classification dataset, how many one-vs-rest 2 class problems can be defined for this dataset?

a) 4
b) 5
c) 6
d) 8
e) 16

F. Normal Distribution

7. What is the square of standard deviation?
   a) mean
   b) variance
   c) gauss
   d) bias

8. When the covariance of two variables are zero and their variances are different, what is the shape of the isoprobability contour plot?
   a) square
   b) ellipse
   c) circle
   d) ellipse with 45 degree

G. Multivariate Classification

9. Which of the following classifiers is not linear?
   a) Qda
   b) Lda
   c) Nearest Mean
   d) Naive Bayes

H. Support Vector Machines

10. Which of the following is the function to be optimized in support vector classification?
   a) min $R^2 + C\sum_t \epsilon^t$
   b) min $\frac{1}{2}||w||^2 + C\sum_t(\epsilon_+^t + \epsilon_-^t)$
   c) min $\frac{1}{2}||w||^2 + C\sum_t \epsilon^t$

11. Which of the following is the function to be optimized in support vector regression?
   a) min $R^2 + C\sum_t \epsilon^t$
   b) min $\frac{1}{2}||w||^2 + C\sum_t(\epsilon_+^t + \epsilon_-^t)$
   c) min $\frac{1}{2}||w||^2 + C\sum_t \epsilon^t$

12. Which of the following is the function to be optimized in one-class kernel machines?
   a) min $R^2 + C\sum_t \epsilon^t$
   b) min $\frac{1}{2}||w||^2 + C\sum_t(\epsilon_+^t + \epsilon_-^t)$
   c) min $\frac{1}{2}||w||^2 + C\sum_t \epsilon^t$

13. What are the class labels in Support Vector machine classification?
   a) 0, 1
   b) -1, +1
   c) -1, 0

14. What is the property that must be satisfied by the support vectors in terms of α^t for classification in separable case?
   a) $\alpha^t = 0$
   b) $\alpha^t < 0$
   c) $\alpha^t > 0$

I. Decision Trees

Consider learning a decision tree that you could use to judge whether or not you will like a given restaurant. Assume you have chosen to use the following two features to describe restaurants, with the possible values shown.

Price $\in$ {Low, Med, High} Type $\in$ {Hamburgers, Pizza, Fish, Vegetarian}

Assume decision algorithm is given the following set of classified training examples.

P	T	Like
L	H	+
L	V	+
M	F	-
M	V	+
H	P	-

$\log 2 = 1, \log 3 = 1.58, \log 4 = 2, \log 5 = 2.32.$

15. Which of the following feature will be selected feature at the root node?
   a) Price
   b) Type

Assume you are given the following three features of TV shows, with the possible values shown, and wish to learn how to predict future top-10 shows.

$$Type \in \{Comedy, Drama, News, Sports\}$$

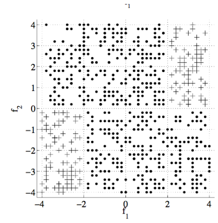
$$Location \in \{LA, NYC, Various\}$$

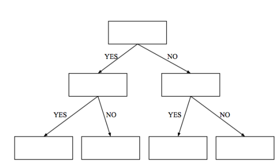
$$Duration \in \mathcal{Z}$$

Assume decision tree construction algorithm is used on the following examples.

Type	Location	Duration	ClassInfo
C	NYC	30	+
C	LA	40	-
N	Var	60	+
N	LA	60	+
S	Var	40	-
D	LA	40	-

16. Which of the following feature will be selected feature at the root node?
   a) Type
   b) Location
   c) Duration





17. What will be the first three nodes of the decision tree that will be constructed for the data above?

a) $f_1 > 2, f_2 > 0, f_1 > -2$
b) $f_1 > 2, f_1 > -2, f_2 > 0$
c) $f_1 > -2, f_2 > 0, f_1 > 2$
d) $f_1 > -2, f_1 > 2, f_2 > 0$
e) $f_2 > 0, f_1 > 2, f_1 > -2$

J. Linear Discrimination

18. Which type of function will not result in a nonlinear function when used as a basis function in linear discrimination?

a) sin
b) exp
c) linear
d) log

19. How many separate hyperplanes (discriminant functions) are defined when K = 10 in linear classification?

a) 2
b) 5
c) 9
d) 10
e) 20

20. How many separate hyperplanes (discriminant functions) are defined when K = 10 in pairwise linear classification?

a) 2
b) 5
c) 9
d) 10
e) 45

21. Which of the following is the sigmoid function?

a) $\frac{1}{x}$
b) $\frac{1}{lnx}$
c) $\frac{1}{1+lnx}$
d) $\frac{1}{1+e^{-x}}$
e) $\frac{1}{1+e^{x}}$

22. What is the value of sigmoid function when x = ∞ and x = -∞?

a) 1, 0
b) 0, 1
c) -1, 1
d) 1, -1
e) -1, 0

23. What is the derivative of the sigmoid function y?

a) y
b) 1 - y
c) y (1 - y)
d) ln y
e) $\frac{1}{y}$

24. Which of the following is the softmax function?

a) $\frac{1}{x}$
b) $\frac{e^x}{\sum_j^K e^j}$
c) $\frac{1}{1+lnx}$
d) $\frac{1}{1+e^{-x}}$
e) $\frac{1}{1+e^{x}}$

25. If there are 3 classes, which of the following vector can be the output of the softmax function?

a) [0.3 0.3 0.3]
b) [0.3 0.3 0.4]
c) [0.4 0.4 0.4]
d) [1.0 1.0 1.0]
e) [0.5 0.5 0.5]

K. Neural Networks

26. How many input and output units (total) are there for a linear perceptron with d = 5 for binary classification?

a) 4
b) 5
c) 6
d) 7
e) 8

27. How many input and output units (total) are there for a linear perceptron with d = 5 for K = 4 classification?

a) 8
b) 9
c) 10
d) 15
e) 24

28. Which of the following functions can not be learned by linear perceptron?

a) and
b) or
c) xor
d) not

29. How many input and output units (total) are there for a multi layer perceptron with d = 5, hidden node count 3, for binary classification?

a) 20
b) 22
c) 26
d) 30
e) 34

30. How many input and output units (total) are there for a linear perceptron with d = 5, hidden node count 3, for K = 4 classification?

a) 20
b) 22
c) 26
d) 30
e) 34